

f(ind)

{ if (ind == 0) return 0;

minSteps = INT_MAX;

for (j = 1; j <= k; j++)

{ if (ind - j >= 0) jump = f(ind - j) + abs(a[ind] - a[ind - j]);
minSteps = min(minSteps, jump);
}



}
return minSteps;

Memoization

→ dp[n]



2.50

```

for (i = 1; i < n; i++)
{
    minSteps = INT_MAX;
    for (j = 1; j <= k; j++)
    {
        if (i - j >= 0)
        {
            jump = dp[i - j] + abs(a[i] - a[i - j]);
            minSteps = min(minSteps, jump);
        }
    }
}

```

$dp[i] = minSteps;$

$T.C \rightarrow O(N \times K)$

